Becca Bond

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Seeking a mechanical engineering internship starting summer 2027 *Fall 2026 coursework in italics*

Education

Olin College of Engineering, *Needham, MA,* B.S. in Mechanical Engineering, May 2029

Relevant Coursework: Calculus BC, Linear Algebra, Multivariable Calculus, Modeling & Simulation, Mechanical Prototyping, Instrumentation & Measurement, *Principles of Integrated Engineering, Materials Science, Calc III, Thermodynamics*

Skills

Software: SolidWorks, OnShape, MATLAB, Python, PrusaSlicer, GrabCAD

Tools: Band saw, Drill press, Angle grinder, Compound miter saw, Rotary saw, Table saw, Impact driver, Palm sander, Belt sander, Soldering iron, 3D printers (MK4S, MK3S, MINI, CORE One, Stratasys f170 & f190), CNC plasma cutter, Abrasive blast cabinet, TIG welder, Wood lathe, Metal lathe, Manual metal mill, Tormach CNC mill, Sheet metal tools, Spot welder

Projects, Experience and Interests

Quad Tailsitter VTOL (Olin College Aerospace team project, Fall 2025-Spring 2026)

- Worked on a team to design and construct a quad tailsitter VTOL for the 2026 Student Unmanned Aerial Systems (SUAS) competition; collaborated with a subteammate to design, optimize and fabricate motor mounts and junctions.
- Self-taught CAD through design of a complex part; incorporated thrust orientation, accessibility and aerodynamics.

Automatic Trebuchet (Olin Mechanical Prototyping team project, Spring 2026)

- Designed & fabricated automatic trebuchet requiring machined parts and motion-converting mechanisms; devised original winch-and-release mechanism using ball-spline shaft, wedge, and systematically engaging/disengaging spool.

Scientific Eco-Monitoring Rover, "Sonic" (Personal team project under Olin CREST Lab, Spring 2026)

- Integrated Raspi motor control, GPS positioning and sensor monitoring into terrain rover for field scientific support.

Electric Go Kart (Personal team project, Fall 2025)

- Worked with a partner to reconstruct an electric go kart: replaced battery, motor, boost converter, motor controller and sprockets; manufactured mounts, axle keys and spacers; installed lights, power switch and boost converter.
- Practiced fundamentals of electric circuits; optimized speed & acceleration based on converted power & gear ratio.

F450 Quadcopter (Embry-Riddle UAS Camp, Summer 2025)

- Collaborated on a team of students to build, configure and operate an F450 Quadcopter as well as three foam gliders.
- Learned and applied fundamentals of UAS (Unmanned Aircraft Systems) manufacturing, technology and operation.

Convertible Mobile Vending Cart (Personal project, Summer 2025)

- Designed a mobile cart rapidly transformable into a vending booth; used cart to sell snocones as a youth hockey fund.
- Creatively used common materials (scrap wood, swivel chair base plate, bike wheels, broom) to construct booth supports, wheel-locking mechanism, and cooler & generator storage; functionalized two-minute setup & teardown.

Wheelhouse Sports Complex (Volunteering team project, Summer 2024)

- Constructed parts of a hockey rink, volleyball court, boxing ring, lockers and offices; assisted with the facility's structural assembly & critical path tasks, plus team organization & planning.

Mechanical Archer (Personal project, Summer 2023)

- Created mechanical arming & firing system for a bow-and-arrow: devised spring mechanism, trigger operation, and arm joint; rapidly prototyped archer using repurposed materials (scrap wood, metal fasteners & spring, paint bucket).

Activities: Olin College aerospace team, National youth Bible quizzing, Professional inline hockey, *Olin Shop assistant (F '26)*

Leadership: NYI youth Bible quizzing – Team captain, Coach, Student tutor & mentor

Interests: Mechanical engineering; Aerospace engineering; Multidisciplinary robotics; Manufacturing; Everything fish